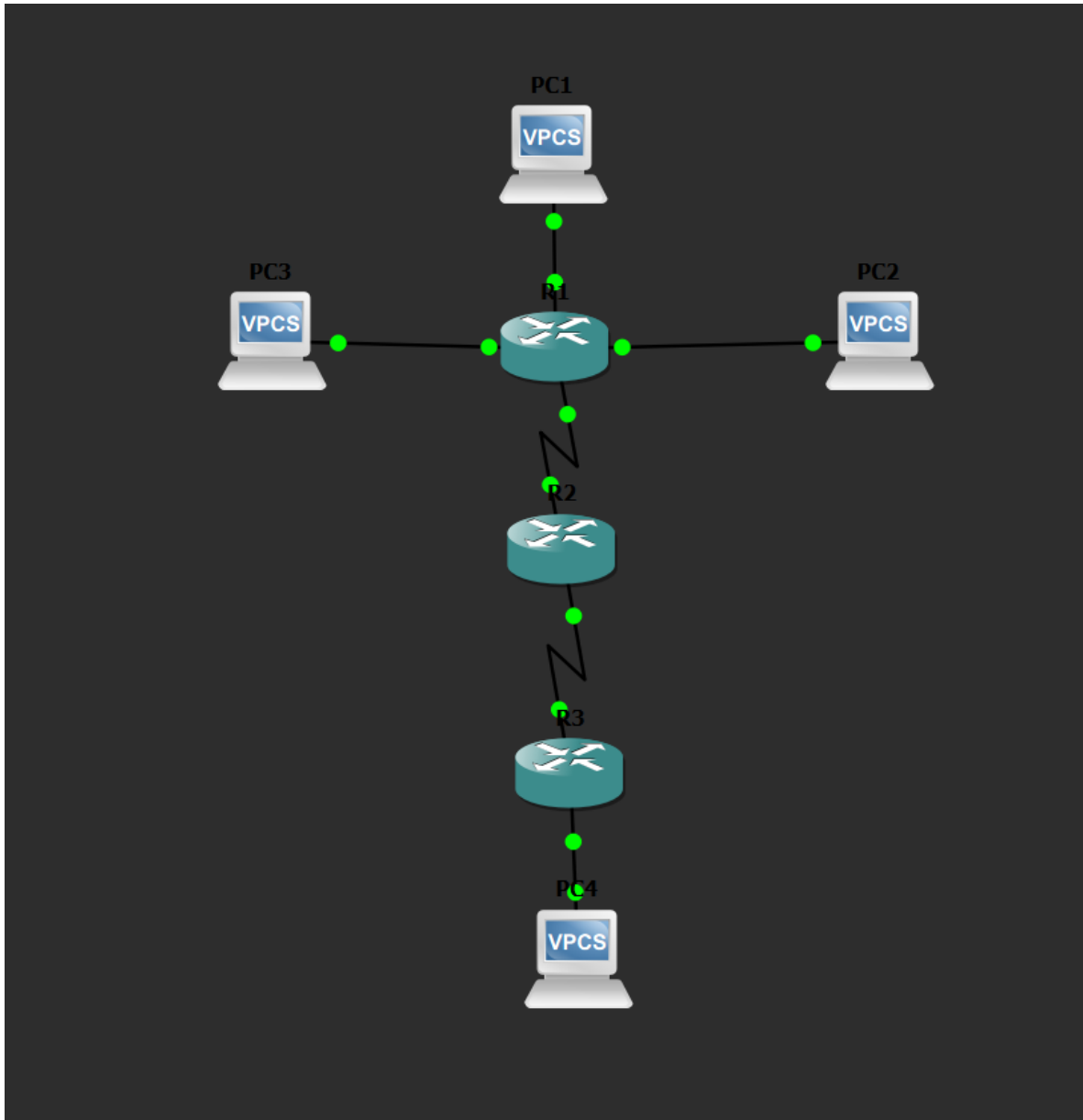


Ejercicio 1

Topología implementada en GNS3.



- Configuración IP de las PCs.

```
PC1> ip 10.4.24.2 /26 10.4.24.1
Checking for duplicate address...
save
PC1 : 10.4.24.2 255.255.255.192 gateway 10.4.24.1

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 10.4.24.66 /27 10.4.24.65
Checking for duplicate address...
savePC1 : 10.4.24.66 255.255.255.224 gateway 10.4.24.65

PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC3>
PC3> ip 10.4.24.98 /28 10.4.24.97
Checking for duplicate address...
PC1 : 10.4.24.98 255.255.255.240 gateway 10.4.24.97

PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> ip 10.4.24.114 /29 10.4.24.113
Checking for duplicate address...
savePC1 : 10.4.24.114 255.255.255.248 gateway 10.4.24.113

PC4> save
Saving startup configuration to startup.vpc
. done
```

- Configuración de interfaces en routers.

```
interface FastEthernet0/0
 ip address 10.4.24.1 255.255.255.192
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 10.4.24.65 255.255.255.224
 duplex auto
 speed auto
!
interface Serial1/0
 ip address 10.4.24.121 255.255.255.252
 serial restart-delay 0
!
```

```
R2(config)#hostname R2
R2(config)#interface Serial 1/0
R2(config-if)#ip address 10.4.24.122 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar  1 00:02:50.015: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R2(config)#
*Mar  1 00:02:51.019: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0,
  changed state to up
R2(config)#interface Serial 1/1
R2(config-if)#ip address 10.4.24.125 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar  1 00:03:07.431: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R2(config)#
*Mar  1 00:03:08.435: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1,
  changed state to up
R2(config)#ip route 10.4.24.0 255.255.255.192 10.4.24.121
R2(config)#ip route 10.4.24.64 255.255.255.224 10.4.24.121
R2(config)#ip route 10.4.24.96 255.255.255.240 10.4.24.121
R2(config)#ip route 10.4.24.112 255.255.255.248 10.4.24.126
```

```

interface FastEthernet2/0
 ip address 10.4.24.97 255.255.255.240
 duplex auto
 speed auto
!
ip forward-protocol nd
ip route 10.4.24.112 255.255.255.248 10.4.24.122
ip route 10.4.24.124 255.255.255.252 10.4.24.122

```

```

R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname R3
R3(config)#interface FastEthernet 0/0
R3(config-if)#ip address 10.4.24.113 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:05:23.627: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:05:24.627: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R3(config)#interface Serial 1/0
R3(config-if)#ip address 10.4.24.126 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:05:37.895: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R3(config)#
*Mar 1 00:05:38.899: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0,
changed state to up
R3(config)#ip route 10.4.24.0 255.255.255.192 10.4.24.125
R3(config)#ip route 10.4.24.64 255.255.255.224 10.4.24.125
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#ip route 10.4.24.96 255.255.255.240 10.4.24.125
R3(config)#ip route 10.4.24.120 255.255.255.252 10.4.24.125
R3(config)#end

```

- **Tabla de rutas estáticas configuradas.**

```

R2(config)#ip route 10.4.24.0 255.255.255.192 10.4.24.121
R2(config)#ip route 10.4.24.64 255.255.255.224 10.4.24.121
R2(config)#ip route 10.4.24.96 255.255.255.240 10.4.24.121
R2(config)#ip route 10.4.24.112 255.255.255.248 10.4.24.126
R2(config)#

```

```

R3(config)#ip route 10.4.24.0 255.255.255.192 10.4.24.125
R3(config)#ip route 10.4.24.64 255.255.255.224 10.4.24.125
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#ip route 10.4.24.96 255.255.255.240 10.4.24.125
R3(config)#ip route 10.4.24.120 255.255.255.252 10.4.24.125

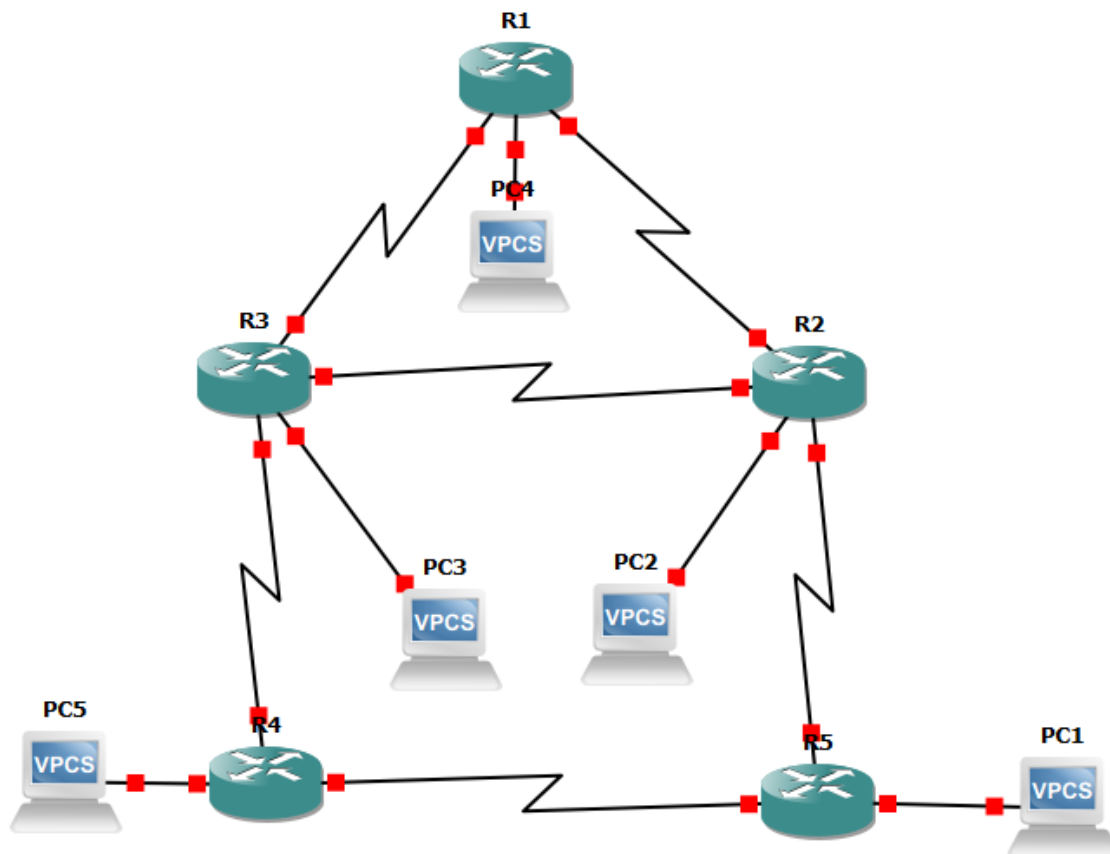
```

- Pruebas de ping exitosas entre las PCs indicadas.

```
ping 10.4.24.114
10.4.24.114 icmp_seq=1 timeout
84 bytes from 10.4.24.114 icmp_seq=2 ttl=61 time=92.890 ms
84 bytes from 10.4.24.114 icmp_seq=3 ttl=61 time=92.963 ms
84 bytes from 10.4.24.114 icmp_seq=4 ttl=61 time=91.528 ms
84 bytes from 10.4.24.114 icmp_seq=5 ttl=61 time=93.647 ms
```

Ejercicio 2

Topología implementada en GNS3.



- Configuración IP de las PCs.

```
PC1> ip 192.4.24.2 /26 192.4.24.1
Checking for duplicate address...
PC1 : 192.4.24.2 255.255.255.192 gateway 192.4.24.1

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 192.4.24.66 /27 192.4.24.65
Checking for duplicate address...
PC1 : 192.4.24.66 255.255.255.224 gateway 192.4.24.65

PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> ip 192.4.24.98 /28 192.4.24.97
Checking for duplicate address...
PC1 : 192.4.24.98 255.255.255.240 gateway 192.4.24.97

PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> ip 192.4.24.114 /28 192.4.24.113
Checking for duplicate address...
saPC1 : 192.4.24.114 255.255.255.240 gateway 192.4.24.113

PC4> save
Saving startup configuration to startup.vpc
. done
```

```
PC5> ip 192.4.24.130 /29 192.4.24.129
Checking for duplicate address...
PC1 : 192.4.24.130 255.255.255.248 gateway 192.4.24.129

PC5> save
Saving startup configuration to startup.vpc
. done
```

- **Configuración de interfaces en routers.**

```
R1
R1(config)#interface FastEthernet 0/0
R1(config-if)#ip address 192.4.24.113 255.255.255.240
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:01:09.683: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:01:10.683: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#interface Serial 1/0
R1(config-if)#ip address 192.4.24.137 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:01:22.759: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config)#
*Mar 1 00:01:23.763: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config)#interface Serial 1/1
R1(config-if)#
*Mar 1 00:01:44.123: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-if)#ip address 192.4.24.141 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:25.103: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R1(config)#
*Mar 1 00:02:26.107: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R1(config)#network 192.4.24.112 0.0.0.15 area 0
^
% Invalid input detected at '^' marker.

R1(config)#router ospf 1
R1(config-router)#network 192.4.24.112 0.0.0.15 area 0
R1(config-router)#network 192.4.24.136 0.0.0.3 area 0
R1(config-router)#
*Mar 1 00:02:54.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
R1(config-router)#network 192.4.24.140 0.0.0.3 area 0
R1(config-router)#
```

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```
R2
R2(config)#interface FastEthernet 0/0
R2(config-if)#ip address 192.4.24.65 255.255.255.224
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:02:34.535: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:35.535: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#ip address 192.4.24.142 255.255.255.252 no shutdown
^
% Invalid input detected at '^' marker.

R2(config)#interface Serial 1/1
R2(config-if)#ip address 192.4.24.142 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:10.099: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R2(config)#
*Mar 1 00:03:11.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R2(config)#interface Serial 1/2
R2(config-if)#ip address 192.4.24.146 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:27.035: %LINK-3-UPDOWN: Interface Serial1/2, changed state to up
R2(config)#
*Mar 1 00:03:28.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to up
R2(config)#interface Serial 1/3
R2(config-if)#ip address 192.4.24.158 255.255.255.252
R2(config-if)#
*Mar 1 00:03:54.155: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to down
R2(config-if)#no shutdown
R2(config-if)#exit
```



```
R2(config)#router ospf 1
R2(config-router)#network 192.4.24.64 0.0.0.31 area 0
R2(config-router)#network 192.4.24.140 0.0.0.3 area 0
R2(config-router)#
*Mar 1 00:05:17.947: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.141 on Serial1/1 from LOADING to FULL, Loading Done
R2(config-router)#network 192.4.24.144 0.0.0.3 area 0
R2(config-router)#network 192.4.24.156 0.0.0.3 area 0

R3(config)#interface FastEthernet 0/0
R3(config-if)#ip address 192.4.24.97 255.255.255.240
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:05:19.707: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:05:20.707: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#interface Serial 1/0
R3(config-if)#ip address 192.4.24.138 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:05:32.139: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R3(config)#
*Mar 1 00:05:33.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R3(config)#interface Serial 1/2
R3(config-if)#ip address 192.4.24.145 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit

R3(config)#interface Serial 1/3
R3(config-if)#ip address 192.4.24.149 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:06:52.695: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R3(config)#
*Mar 1 00:06:53.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R3(config)#router ospf 1
R3(config-router)#network 192.4.24.96 0.0.0.15 area 0
R3(config-router)#
R3(config-router)#
R3(config-router)#
*Mar 1 00:07:14.303: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
R3(config-router)#network 192.4.24.136 0.0.0.3 area 0
R3(config-router)#
*Mar 1 00:07:23.211: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.141 on Serial1/0 from LOADING to FULL, Loading Done
R3(config-router)#network 192.4.24.144 0.0.0.3 area 0
R3(config-router)#
*Mar 1 00:07:27.983: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.158 on Serial1/2 from LOADING to FULL, Loading Done
R3(config-router)#network 192.4.24.148 0.0.0.3 area 0
```

```
R5
R5(config)#hostname R5
R5(config)#interface FastEthernet 0/0
R5(config-if)#ip address 192.4.24.1 255.255.255.192
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:09:51.551: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:09:52.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5(config)#interface Serial 1/0
R5(config-if)#ip address 192.4.24.154 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:10:05.215: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R5(config)#
*Mar 1 00:10:06.219: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R5(config)#interface Serial 1/3
R5(config-if)#ip address 192.4.24.157 255.255.255.252
R5(config-if)#NT
% Incomplete command.

R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:10:24.979: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R5(config)#
*Mar 1 00:10:25.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R5(config)#router ospf 1
R5(config-router)#network 192.4.24.0 0.0.0.63 area 0
R5(config-router)#network 192.4.24.152 0.0.0.3 area 0
R5(config-router)#
*Mar 1 00:10:56.975: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.153 on Serial1/0 from LOADING to FULL, Loading Done
R5(config-router)#network 192.4.24.156 0.0.0.3 area 0
R5(config-router)#
*Mar 1 00:11:05.379: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.158 on Serial1/3 from LOADING to FULL, Loading Done
R5(config-router)#
```

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```
R4
R4(config)#interface FastEthernet 0/0
R4(config-if)#ip address 192.4.24.129 255.255.255.248
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:06:45.479: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:46.479: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#ip address 192.4.24.153 255.255.255.252
^
% Invalid input detected at '^' marker.

R4(config)#interface Serial 1/0
R4(config-if)#ip address 192.4.24.153 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:07:06.259: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R4(config)#
*Mar 1 00:07:07.263: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R4(config)#interface Serial 1/3
R4(config-if)#ip address 192.4.24.150 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
*Mar 1 00:07:20.671: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R4(config-if)#exit
R4(config)#
*Mar 1 00:07:21.675: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R4(config)#router ospf 1
R4(config-router)#network 192.4.24.128 0.0.0.7 area 0
R4(config-router)#network 192.4.24.148 0.0.0.3 area 0
R4(config-router)#
*Mar 1 00:07:33.903: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R4(config-router)#
*Mar 1 00:07:35.955: %OSPF-5-ADJCHG: Process 1, Nbr 192.4.24.149 on Serial1/3 from LOADING to FULL, Loading Done
R4(config-router)#network 192.4.24.152 0.0.0.3 area 0
R4(config-router)#
```

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- **Tabla de rutas estáticas configuradas.**

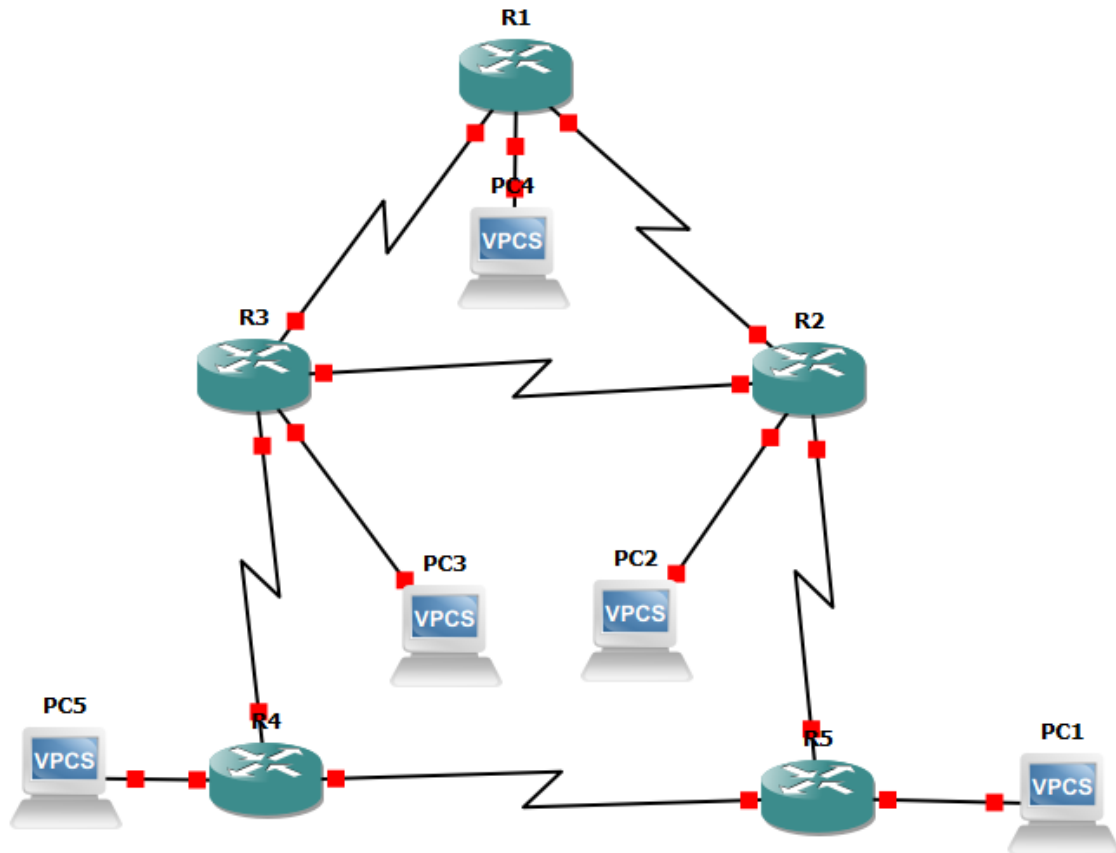
```
R1#show ip route ospf
    192.4.24.0/24 is variably subnetted, 11 subnets, 5 masks
O        192.4.24.156/30 [110/128] via 192.4.24.142, 00:05:42, Serial1/1
O        192.4.24.152/30 [110/192] via 192.4.24.142, 00:05:32, Serial1/1
          [110/192] via 192.4.24.138, 00:06:12, Serial1/0
O        192.4.24.148/30 [110/128] via 192.4.24.138, 00:13:32, Serial1/0
O        192.4.24.144/30 [110/128] via 192.4.24.142, 00:17:22, Serial1/1
          [110/128] via 192.4.24.138, 00:15:43, Serial1/0
O        192.4.24.128/29 [110/138] via 192.4.24.138, 00:13:12, Serial1/0
O        192.4.24.64/27 [110/74] via 192.4.24.142, 00:21:19, Serial1/1
O        192.4.24.96/28 [110/74] via 192.4.24.138, 00:15:53, Serial1/0
O        192.4.24.0/26 [110/138] via 192.4.24.142, 00:05:32, Serial1/1
```

- **Pruebas de ping exitosas entre las PCs indicadas.**

```
ping 192.4.24.2
192.4.24.2 icmp_seq=1 timeout
84 bytes from 192.4.24.2 icmp_seq=2 ttl=62 time=61.643 ms
84 bytes from 192.4.24.2 icmp_seq=3 ttl=62 time=61.645 ms
84 bytes from 192.4.24.2 icmp_seq=4 ttl=62 time=61.368 ms
84 bytes from 192.4.24.2 icmp_seq=5 ttl=62 time=61.045 ms
```

Ejercicio 3

Topología implementada en GNS3.



- Configuración IP de las PCs.

```
PC1> ip 172.16.4.2 /26 172.16.4.1
Checking for duplicate address...
PC1 : 172.16.4.2 255.255.255.192 gateway 172.16.4.1

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 172.16.4.66 /27 172.16.4.65
Checking for duplicate address...
PC1 : 172.16.4.66 255.255.255.224 gateway 172.16.4.65

PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> ip 172.16.4.98 /28 172.16.4.97
Checking for duplicate address...
saPC1 : 172.16.4.98 255.255.255.240 gateway 172.16.4.97

PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> ip 172.16.4.114 /28 172.16.4.113
Checking for duplicate address...
PC1 : 172.16.4.114 255.255.255.240 gateway 172.16.4.113

PC4> svae
Bad command: "svae". Use ? for help.

PC4> save
Saving startup configuration to startup.vpc
. done
```

```
PC5> ip 172.16.4.130 /29 172.16.4.129
Checking for duplicate address...
savePC1 : 172.16.4.130 255.255.255.248 gateway 172.16.4.129

PC5> save
Saving startup configuration to startup.vpc
. done
```

- Configuración de interfaces en routers.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#interface FastEthernet 0/0
R1(config-if)#ip address 172.16.4.113 255.255.255.240
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:01:44.767: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:01:45.799: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#interface Serial 1/0
R1(config-if)#ip address 172.16.4.137 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:01:59.279: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config)#
*Mar 1 00:02:00.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config)#interface Serial 1/1
R1(config-if)#ip address 172.16.4.141 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:17.259: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R1(config)#
*Mar 1 00:02:18.263: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R1(config)#router ospf 1
R1(config-router)#
*Mar 1 00:02:24.055: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-router)#network 172.16.4.112 0.0.0.15 area 0
R1(config-router)#network 172.16.4.136 0.0.0.3 area 0
R1(config-router)#network 172.16.4.140 0.0.0.3 area 0
R1(config-router)#
*Mar 1 00:02:44.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
R1(config-router)#
```

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```
R2(config)#router ospf 1
R2(config-router)#network 172.16.4.64 0.0.0.31 area 0
R2(config-router)#network 172.16.4.140 0.0.0.3 area 0
R2(config-router)#
*Mar 1 00:05:29.963: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.4.141, State 2, Cost 1
R2(config-router)#network 172.16.4.144 0.0.0.3 area 0
```

```
R2
R2(config)#interface FastEthernet 0/0
R2(config-if)#ip address 172.16.4.65 255.255.255.224
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:46.935: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:47.935: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#interface Serial 1/1
R2(config-if)#ip address 172.16.4.142 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:57.991: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R2(config)#
*Mar 1 00:03:58.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R2(config)#interface Serial 1/2
R2(config-if)#ip address 172.16.4.146 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:04:18.143: %LINK-3-UPDOWN: Interface Serial1/2, changed state to up
R2(config)#
*Mar 1 00:04:19.147: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to up
R2(config)#interface Serial 1/3
R2(config-if)#ip address 172.16.4.157 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#
*Mar 1 00:04:40.615: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R2(config-if)#exit
*Mar 1 00:04:41.619: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R2(config-if)#exit
R2(config)#
*Mar 1 00:04:44.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to down
R2(config)#ip route 172.16.4.0 255.255.255.192 172.16.4.158
R2(config)#ip route 172.16.4.152 255.255.255.252 172.16.4.158
R2(config)#
```

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```
R3(config)#ip route 172.16.4.128 255.255.255.248 172.16.4.150
R3(config)#ip route 172.16.4.152 255.255.255.252 172.16.4.150
R3(config)#
*Mar 1 00:05:34.027: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
R3(config)#router ospf 1
R3(config-router)#network 172.16.4.96 0.0.0.15 area 0
R3(config-router)#network 172.16.4.136 0.0.0.3 area 0
R3(config-router)#
*Mar 1 00:05:47.815: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.4.141 on Serial1/0 from LOADING to FULL, Loading Done
R3(config-router)#network 172.16.4.144 0.0.0.3 area 0
```

```
R3
*Mar 1 00:00:07.855: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state
to administratively down
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface FastEthernet 0/0
R3(config-if)#ip address 172.16.4.97 255.255.255.240
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:04:21.051: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:04:22.051: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#interface Serial 1/0
R3(config-if)#ip address 172.16.4.138 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:04:33.475: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R3(config)#
*Mar 1 00:04:34.479: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R3(config)#interface Serial 1/2
R3(config-if)#ip address 172.16.4.145 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:04:45.991: %LINK-3-UPDOWN: Interface Serial1/2, changed state to up
R3(config)#
*Mar 1 00:04:46.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to up
R3(config)#interface Serial 1/3
R3(config-if)#ip address 172.16.4.149 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:05:03.379: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R3(config)#
*Mar 1 00:05:04.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R3(config)#
```

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```
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#interface FastEthernet 0/0
R4(config-if)#ip address 172.16.4.129 255.255.255.248
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:05:25.523: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:05:26.523: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#interface Serial 1/3
R4(config-if)#ip address 172.16.4.150 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:05:37.343: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
*Mar 1 00:05:38.355: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R4(config)#interface Serial 1/0
R4(config-if)#ip address 172.16.4.153 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:05:52.011: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R4(config)#
*Mar 1 00:05:53.015: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R4(config)#ip route 0.0.0.0 0.0.0.0 172.16.4.149
R4(config)#ip route 172.16.4.0 255.255.255.192 172.16.4.154
R4(config)#
```

```
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#interface FastEthernet 0/0
R5(config-if)#ip address 172.16.4.1 255.255.255.192
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:06:09.071: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:10.127: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5(config)#interface Serial 1/3
R5(config-if)#ip address 172.16.4.158 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:06:20.699: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
R5(config)#
*Mar 1 00:06:21.703: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to up
R5(config)#interface Serial 1/0
R5(config-if)#ip address 172.16.4.154 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:06:32.579: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R5(config)#
*Mar 1 00:06:33.583: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R5(config)#ip route 0.0.0.0 0.0.0.0 172.16.4.157
R5(config)#ip route 172.16.4.128 255.255.255.248 172.16.4.153
R5(config)#
```

• Tabla de rutas estáticas configuradas.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 9 subnets, 5 masks
O E2   172.16.4.152/30 [110/20] via 172.16.4.142, 00:01:13, Serial1/1
        [110/20] via 172.16.4.138, 00:02:54, Serial1/0
O      172.16.4.144/30 [110/128] via 172.16.4.142, 00:05:39, Serial1/1
        [110/128] via 172.16.4.138, 00:04:36, Serial1/0
C      172.16.4.136/30 is directly connected, Serial1/0
C      172.16.4.140/30 is directly connected, Serial1/1
O E2   172.16.4.128/29 [110/20] via 172.16.4.138, 00:02:56, Serial1/0
O E2   172.16.4.0/26 [110/20] via 172.16.4.142, 00:01:15, Serial1/1
C      172.16.4.112/28 is directly connected, FastEthernet0/0
O      172.16.4.96/28 [110/74] via 172.16.4.138, 00:04:38, Serial1/0
O      172.16.4.64/27 [110/74] via 172.16.4.142, 00:08:16, Serial1/1
```

```
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 10 subnets, 5 masks
S      172.16.4.152/30 [1/0] via 172.16.4.158
C      172.16.4.156/30 is directly connected, Serial1/3
C      172.16.4.144/30 is directly connected, Serial1/2
O      172.16.4.136/30 [110/128] via 172.16.4.145, 00:05:11, Serial1/2
        [110/128] via 172.16.4.141, 00:06:44, Serial1/1
C      172.16.4.140/30 is directly connected, Serial1/1
O E2   172.16.4.128/29 [110/20] via 172.16.4.145, 00:03:29, Serial1/2
S      172.16.4.0/26 [1/0] via 172.16.4.158
O      172.16.4.112/28 [110/74] via 172.16.4.141, 00:08:54, Serial1/1
O      172.16.4.96/28 [110/74] via 172.16.4.145, 00:05:13, Serial1/2
C      172.16.4.64/27 is directly connected, FastEthernet0/0
```

```
R4#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is 172.16.4.149 to network 0.0.0.0
```

```
172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks  
C    172.16.4.152/30 is directly connected, Serial1/0  
C    172.16.4.148/30 is directly connected, Serial1/3  
C    172.16.4.128/29 is directly connected, FastEthernet0/0  
S    172.16.4.0/26 [1/0] via 172.16.4.154  
S*  0.0.0.0/0 [1/0] via 172.16.4.149
```

```
R3#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
172.16.0.0/16 is variably subnetted, 10 subnets, 5 masks  
S    172.16.4.152/30 [1/0] via 172.16.4.150  
C    172.16.4.144/30 is directly connected, Serial1/2  
C    172.16.4.148/30 is directly connected, Serial1/3  
C    172.16.4.136/30 is directly connected, Serial1/0  
O    172.16.4.140/30 [110/128] via 172.16.4.146, 00:05:50, Serial1/2  
      [110/128] via 172.16.4.137, 00:05:50, Serial1/0  
S    172.16.4.128/29 [1/0] via 172.16.4.150  
O E2 172.16.4.0/26 [110/20] via 172.16.4.146, 00:02:26, Serial1/2  
O    172.16.4.112/28 [110/74] via 172.16.4.137, 00:05:52, Serial1/0  
C    172.16.4.96/28 is directly connected, FastEthernet0/0  
O    172.16.4.64/27 [110/74] via 172.16.4.146, 00:05:52, Serial1/2
```

```

R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 172.16.4.157 to network 0.0.0.0

    172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.4.152/30 is directly connected, Serial1/0
C       172.16.4.156/30 is directly connected, Serial1/3
S       172.16.4.128/29 [1/0] via 172.16.4.153
C       172.16.4.0/26 is directly connected, FastEthernet0/0
S*    0.0.0.0/0 [1/0] via 172.16.4.157

```

- Pruebas de ping exitosas entre las PCs indicadas.

```

PC5> ping 172.16.4.2
172.16.4.2 icmp_seq=1 timeout
84 bytes from 172.16.4.2 icmp_seq=2 ttl=62 time=61.623 ms
84 bytes from 172.16.4.2 icmp_seq=3 ttl=62 time=61.140 ms
84 bytes from 172.16.4.2 icmp_seq=4 ttl=62 time=61.695 ms
84 bytes from 172.16.4.2 icmp_seq=5 ttl=62 time=61.572 ms

ping 172.16.4.114
172.16.4.114 icmp_seq=1 timeout
84 bytes from 172.16.4.114 icmp_seq=2 ttl=61 time=92.973 ms
84 bytes from 172.16.4.114 icmp_seq=3 ttl=61 time=92.191 ms
84 bytes from 172.16.4.114 icmp_seq=4 ttl=61 time=91.843 ms
84 bytes from 172.16.4.114 icmp_seq=5 ttl=61 time=90.573 ms

```